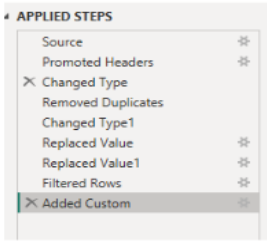
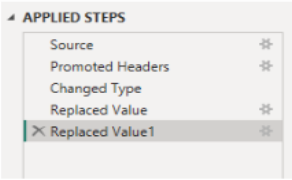
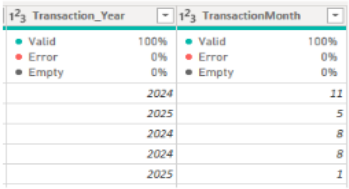
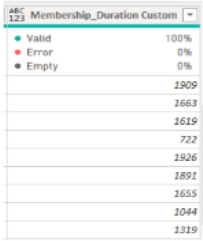
Task 1: Data Modeling and Cleaning

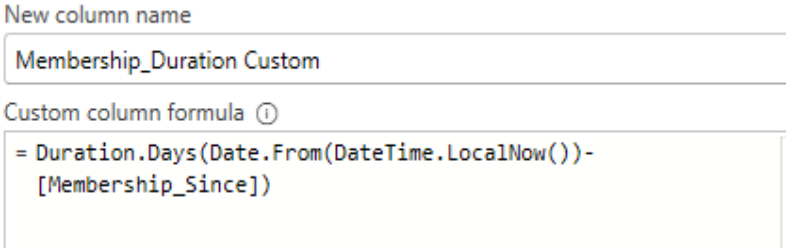




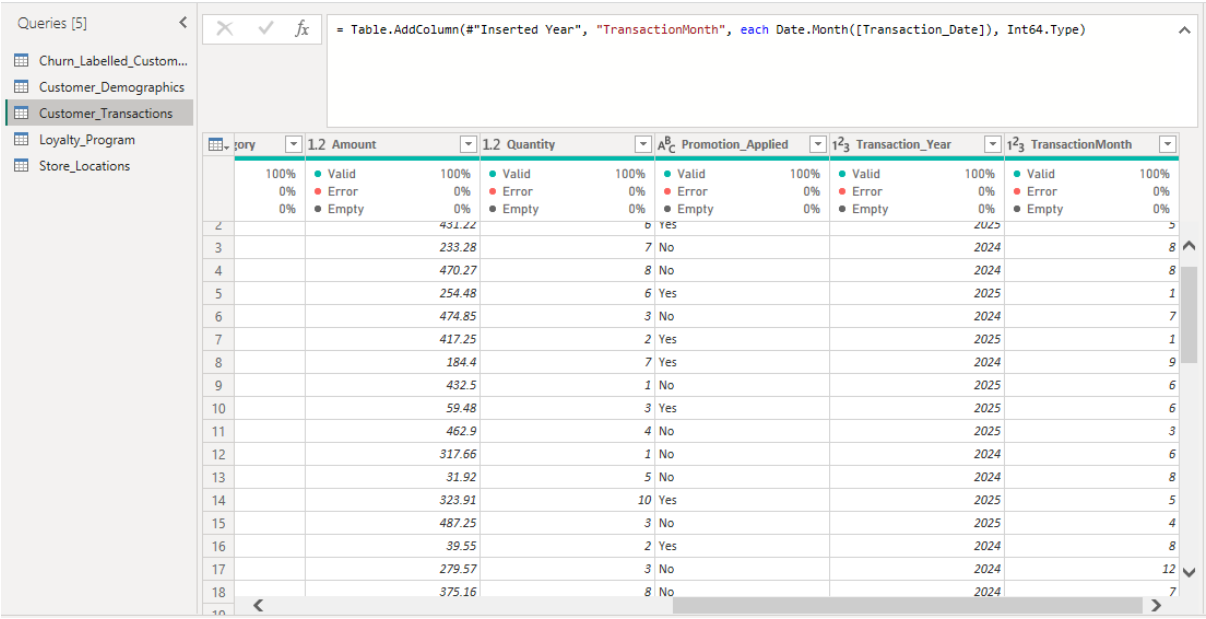




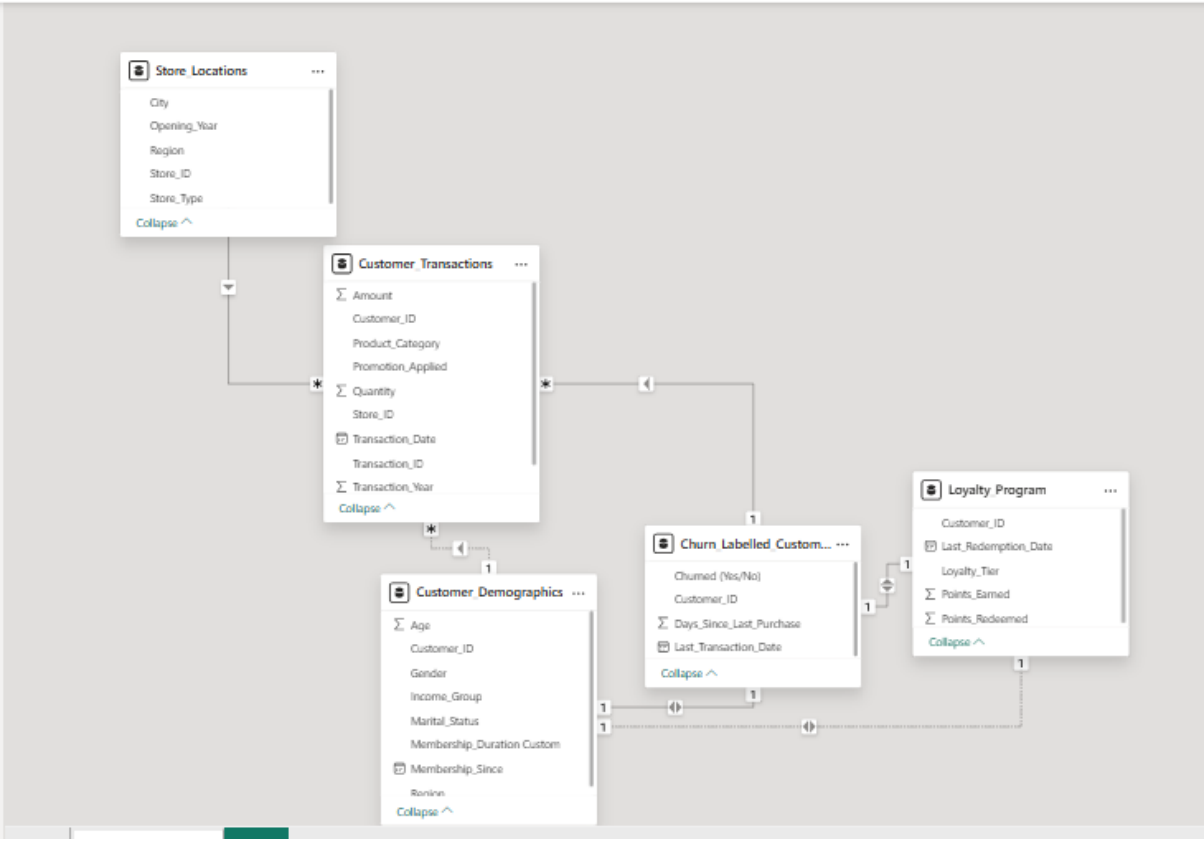
Membership_Duration = Today - Membership_Since



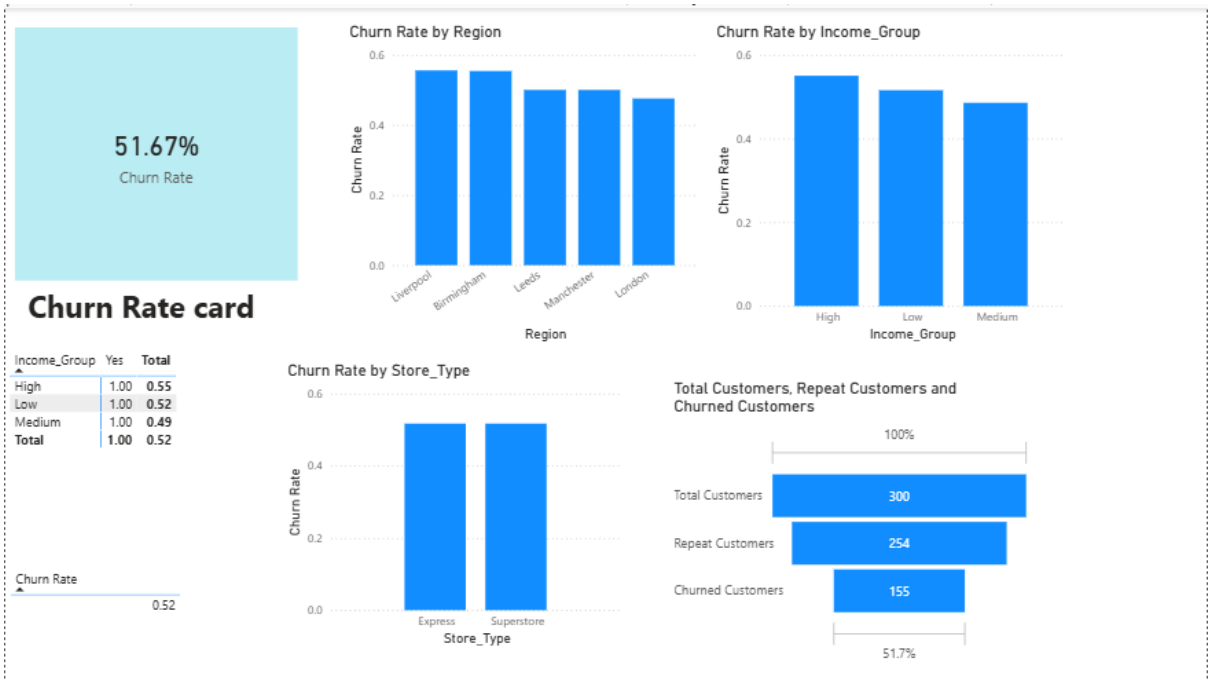
Add a Transaction_Year and Transaction_Month column from Transaction_Date



One to one and many one relationships:



Task 2: Churn and Retention Metrics



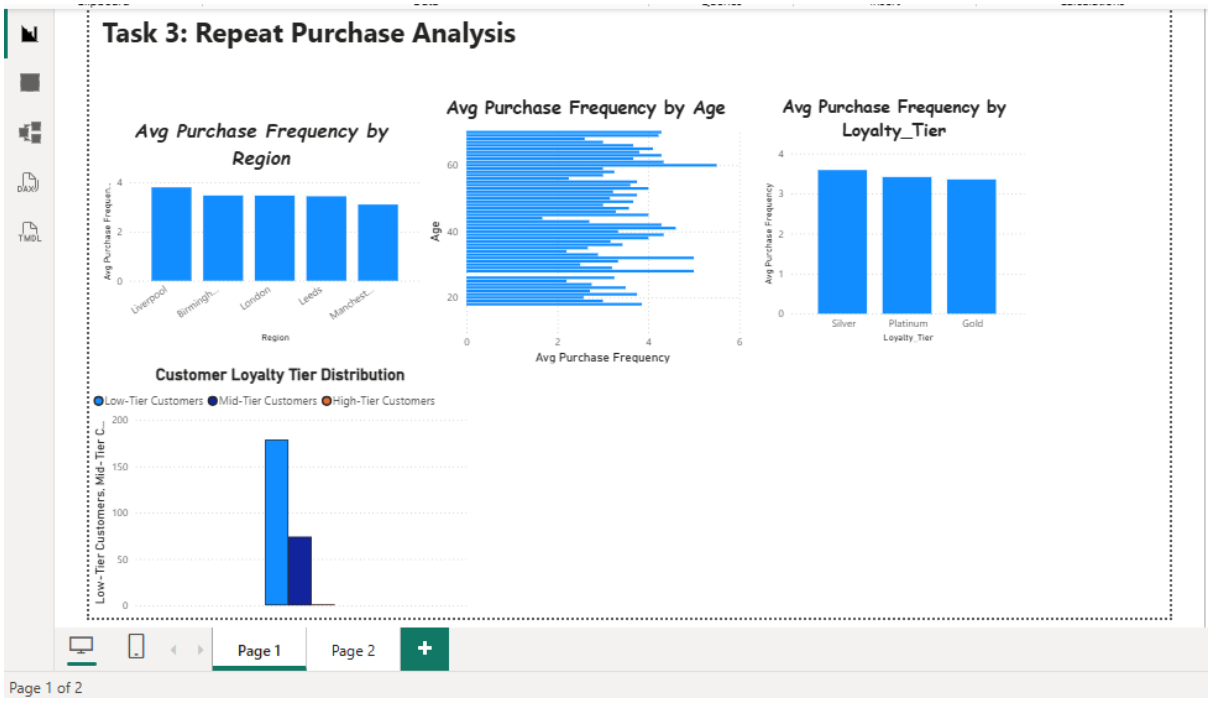
Income_Group	Yes	Total
High	1.00	0.55
Low	1.00	0.52
Medium	1.00	0.49
Total	**1.00**	**0.52**

Churn Rate
0.52

```
Total Customers =
DISTINCTCOUNT(Customer_Demographics[Customer_ID])
```

```
Churn Rate % =
DIVIDE(
    [Churned Customers],
    [Total Customers],
    0
) * 100
```

Task 3: Repeat Purchase Analysis



- Customer_Transactions
  - Amount
  - Avg Purchase Frequency
  - Customer_ID
  - High-Tier Customers
  - Low-Tier Customers
  - Mid-Tier Customers
  - Product_Category
  - Promotion_Applied
  - Quantity
  - Store_ID
  - Total Purchases
  - Transaction_Date
  - Transaction_ID

LOW TIER CUSTOMERS

MID TIER CUSTOMERS

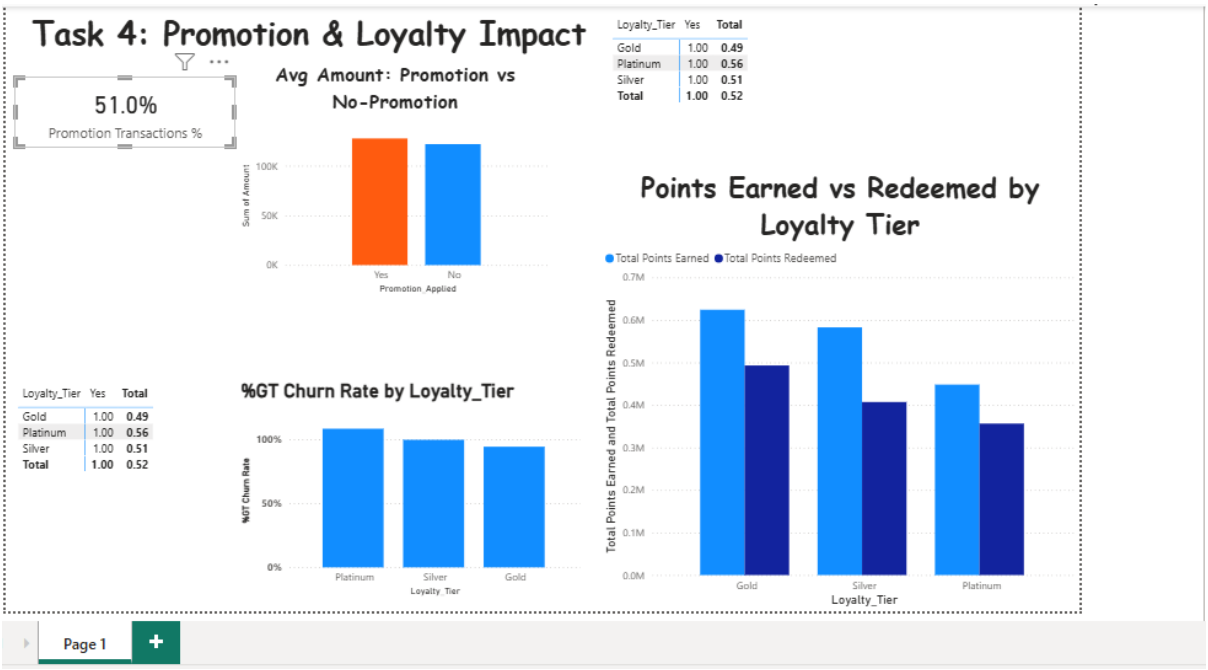
HIGH TIER CUSTOMERS

```
Mid-Tier Customers =
CALCULATE(
    DISTINCTCOUNT(Customer_Transactions[Customer_ID]),
    FILTER(
        VALUES(Customer_Transactions[Customer_ID]),
        [Total Purchases] >= 5 &&
        [Total Purchases] <= 10
    )
)
```

```
High-Tier Customers =
CALCULATE(
    DISTINCTCOUNT(Customer_Transactions[Customer_ID]),
    FILTER(
        VALUES(Customer_Transactions[Customer_ID]),
        [Total Purchases] >= 11
    )
)
```

```
Low-Tier Customers =
CALCULATE(
    DISTINCTCOUNT(Customer_Transactions[Customer_ID]),
    FILTER(
        VALUES(Customer_Transactions[Customer_ID]),
        [Total Purchases] >= 2 &&
        [Total Purchases] <= 4
    )
)
```

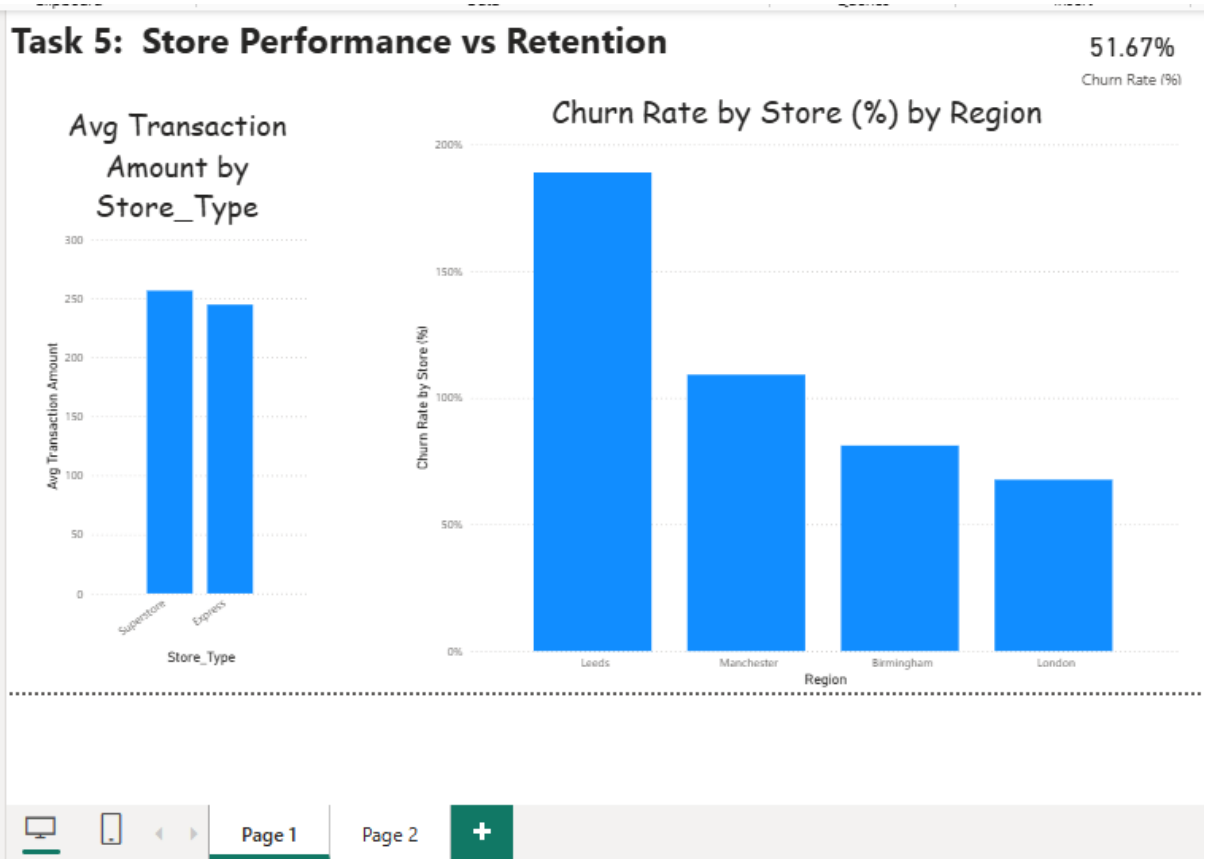
Task 4: Promotion & Loyalty Impact

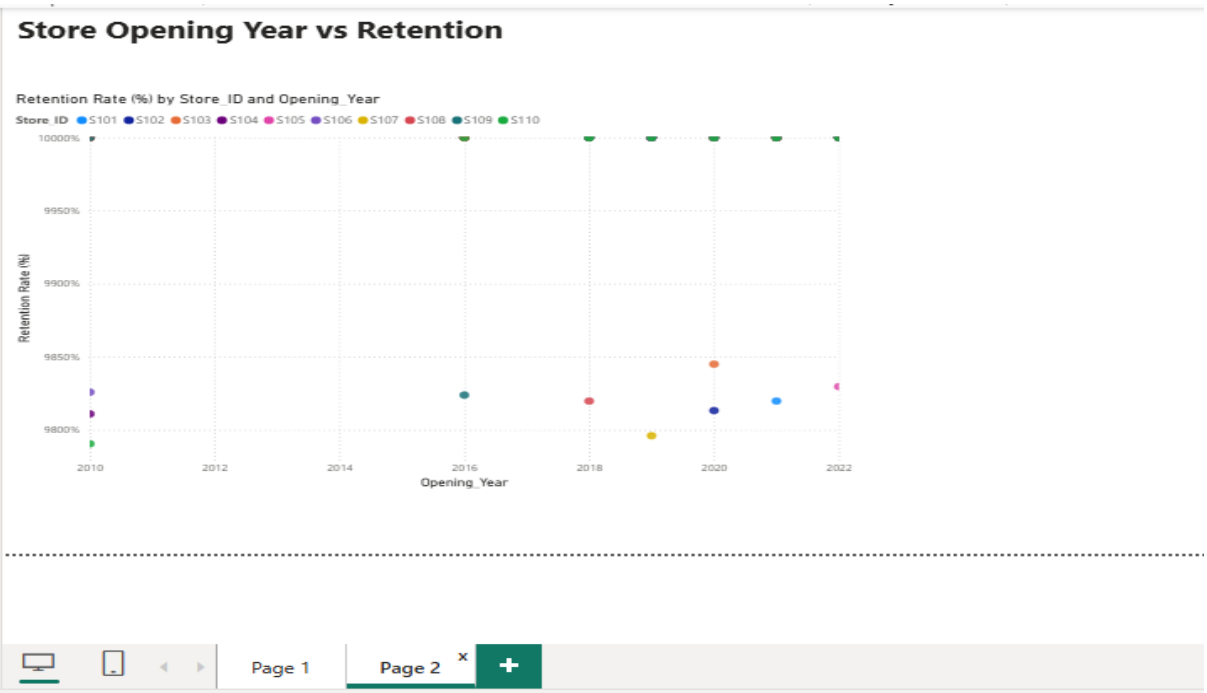



```
Promotion Transactions % =
DIVIDE(
CALCULATE(
COUNTROWS(Customer_Transactions),
Customer_Transactions[Promotion_Applied] = "Yes"
),
COUNTROWS(Customer_Transactions),
0
)
```

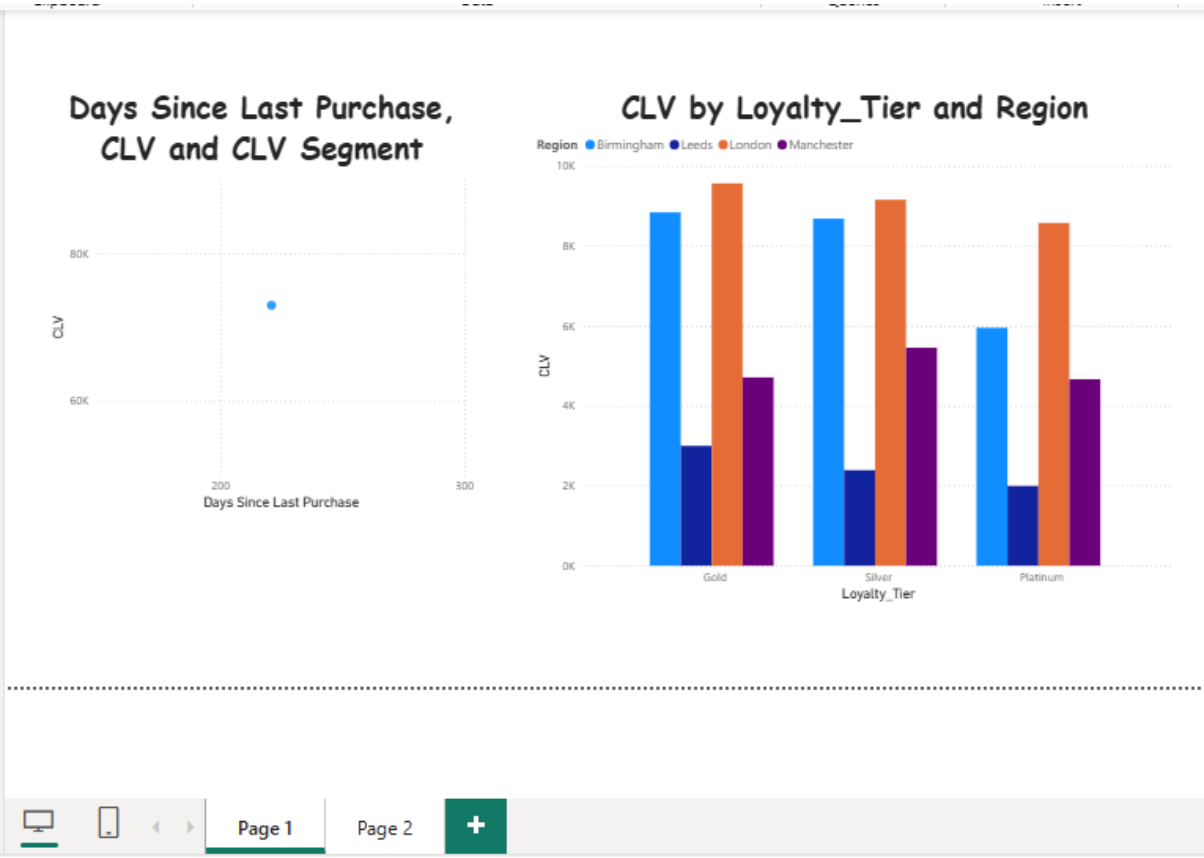
```
Transactions With Promotion =
CALCULATE(
    COUNT(Customer_Transactions[Transaction_ID]),
    Customer_Transactions[Promotion_Applied] = "Yes"
)
```

Task 5: Store Performance vs Retention





Task 6: Customer Value (CLV) Analysis

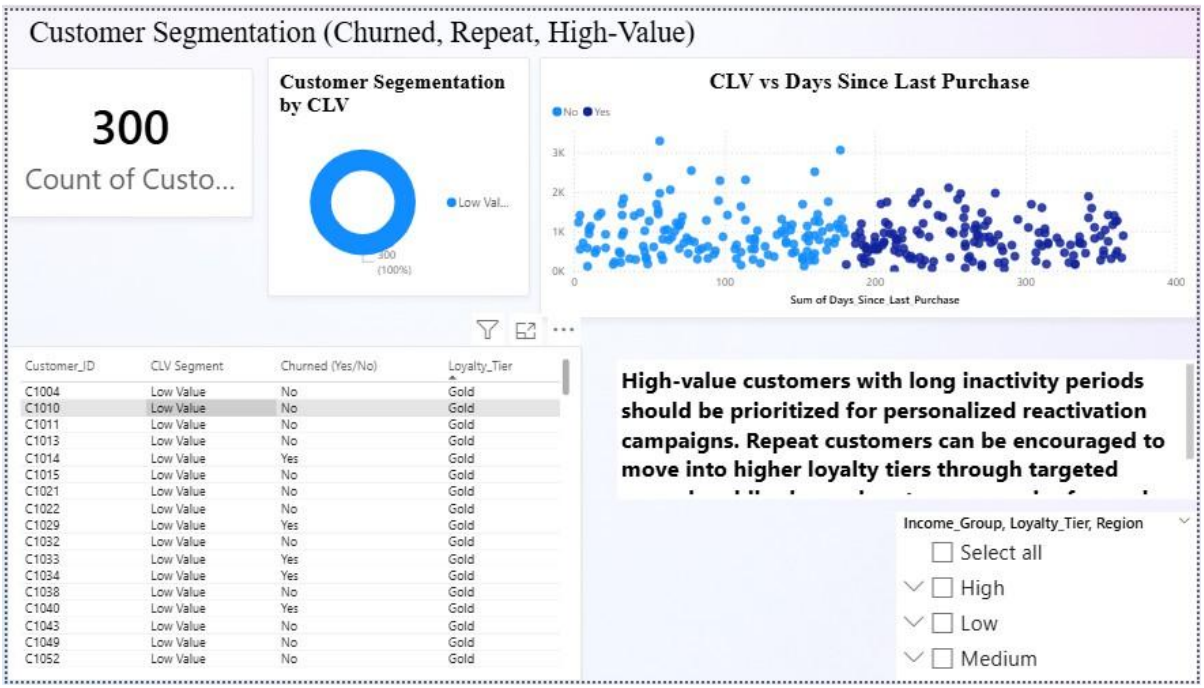


CLV Segment	Count of Customer_ID	CLV
High	1000	72,985.55

```
CLV =
DIVIDE(
    [Total Amount Spent],
    AVERAGE(Customer_Demographics[Membership Duration (Years)])
)
```

```
CLV Segment =
SWITCH(
    TRUE(),
    [CLV] <= PERCENTILEX.INC(Customer_Demographics, [CLV], 0.25), "Low",
    [CLV] <= PERCENTILEX.INC(Customer_Demographics, [CLV], 0.75), "Medium",
    "High"
)
```

Task 7: Final Dashboard and Executive summary



Video Link :
https://drive.google.com/file/d/1J6xoXrTeLS_Pi-_HxPJQeMTQyZSRQvxE/view?usp=sharing